NATIONAL INSTITUTE OF TECHNOLOGY PATNA

Department of Computer Science and Engineering

END SEMESTER EXAMINATION, July- December 2025

B.Tech: CSE Semester-V

Course Name: Internet of Things

Maximum Time: 3 hours **Course Code:** CS54117

Max. Marks: 60

Instruction:

1. Attempt all questions.
2. Assume any suitable data, if necessary. (Any other Instruction need to provide by the concerned faculty)
3. The Marks, CO (Course Outcome) and BL (Bloom's Level) related to questions are mentioned on the right-hand side margin.

		Marks	CO	BL
1.	Issuing Books in a Library involves identifying the member, entering information of the books to be issued, due date, and security check at the gate. Propose an IoT based solution for first issuing of books to the members in a Library. Draw a net diagram of how the components will communicate among each-other and their working procedure. Explain your proposed solution in short.	10	CO4, CO5	E, C
2.	Assume you are connecting a Bluetooth-enabled music system with your smart phone. Describe the steps to establish the connection between the two devices. Analyze which class of Bluetooth will serve your purpose better and optimize the energy consumption.	10	CO3, CO5	C,A
3.	Explain beacon and non-beacon modes of data transmission with diagram for each type.	10	CO3	U, P
4.	Discuss the ZigBee protocol with a net layered diagram.	10	CO3	A, P
5.	You are developing a security system for a smart home framework and decided to use data fusion. What kind of data fusion would you prefer to use. Classify different data fusion techniques available.	10	CO4, CO2	P, E
6.	You are developing a security system for a smart home framework and decided to store data on the cloud. Identify different data acquisition components and stages involved. What are the challenges of data acquisition.	10	CO2	R, U

End of Question

Level	Bloom's Taxonomy
1	Remembering (R)
2	Understanding (U)
3	Applying (P)
4	Analyzing (A)
5	Evaluate (E)
6	Create (C)

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NATIONAL INSTITUTE OF TECHNOLOGY PATNA

Department of Computer Science and Engineering

END SEMESTER SPECIAL EXAMINATION, September 2025

B.Tech: CSE Semester-V

Course Name: Internet of Things **Course Code:** CS54117

Maximum Time: 2 hours **Max. Marks: 30**

Instruction:

1. Attempt all questions.
2. Assume any suitable data, if necessary. (Any other instructions need to provide by the concerned faculty)
3. The Marks, CO (Course Outcome) and BL (Bloom's Level) related to questions are mentioned on the right-hand side margin.

		Marks	CO	BL
1.	We are all very familiar with smartphones and know that a large numbers of sensors and actuators are used to design smartphones. i. Identify five sensors and five actuators present in a smartphone. ii. Describe their applications in the context.	10	CO1	U
2.	There are many IoT communication protocols which are designs based on REST based communication stack. i. List down at least two communication protocols which are designed based on REST. ii. Describe the REST based communication stack for IoT with a layered diagram.	10	CO2	A
3.	i. Describe CoAP message format.	10	CO3	C
	ii. In your opinion for an M2M communication which of the following protocol be chosen: CoAP, HTTP?			
	iii. Justify your answer.			

End of Question

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